PROJECT PROPOSAL · ASSIGNMENT 1 — PROBLEM & DATASET

Tender Risk ML

Predicting problematic public procurements in Kazakhstan from open procurement data

Author
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Repository
github.com/awzwz/tender-risk-ml

Data source
OWS v3 API, goszakup.gov.kz

Task
Binary classification + NLP

1.8M	630,762	≈ 23%	5
procurements per year on goszakup.gov.kz	procurements in the dataset (Q1 2024)	of contracts terminated or never concluded	linked tables: tenders, lots, bids, contracts, RNU

01 Project title

> **Tender Risk ML — predicting problematic public procurements in Kazakhstan from open procurement data**

02 Problem description

Kazakhstan's public procurement system (goszakup.gov.kz) processes roughly **1.8 million procurements per year** — 12.9 million in total as of September 2026, with over 40 million supplier bids — spending a significant share of the state budget.

Every procurement, lot, bid and contract is published through the open **OWS v3 GraphQL API**. Oversight bodies, journalists and auditors, however, work by hand: they can inspect a few hundred procurements a year, while the signs of a problematic procurement — a single bidder, an implausible price discount, a competitive procedure that "fails" and is re-awarded to a single source, a supplier with a history of terminated contracts — are scattered across several linked tables and are invisible at the level of one record.

The practical consequence is that a large share of contracts end badly. In a sample of FY2024 contracts, about **one in five was terminated or never concluded**, and about **7% of procurements were cancelled** after publication. Each such case means a repeated procedure, delayed goods or services, and often wasted budget. Detecting the procurements likely to fail *before* they fail would let a controlling body concentrate its limited attention where it matters.

An earlier engineering project by the author (Tender Radar) approached this with 31 hand-written risk rules. Rules are transparent but arbitrary: their thresholds were chosen by intuition, and there was no way to measure whether a "risky" procurement actually turned out badly. This project replaces intuition with a **supervised model trained on observed outcomes**, using the rules only as candidate features.

03 Project goal

Build and evaluate a machine-learning classifier that, using only information available at the time a procurement is published and awarded, predicts whether the procurement will have a **problematic outcome** — defined as the resulting contract being terminated or never concluded — and identify which characteristics of a procurement are most predictive of that outcome.

04 Project objectives

1 **Dataset construction** A1

Collect a linked snapshot of procurements published in Q1 2024 from the OWS v3 API — procurements, lots, bids, contracts and the register of unreliable suppliers — and publish it as a reproducible, documented dataset (collection script + data dictionary).

2 **Data preparation and EDA** A2

Clean and normalise the data (missing values, duplicates, currency/date formats, text normalisation for Russian/Kazakh lot names), explore distributions and class balance, and define the target rigorously with a leakage-safe **time-based train/validation/test split**.

3 **Feature engineering** A2

Derive tabular features (price, number of bidders, discount, trade method, customer and supplier history, the 31 Tender Radar indicators) and text features (TF-IDF over lot names and descriptions).

4 **Baseline modelling and evaluation** MIDTERM

Train and compare classical models (Logistic Regression, Random Forest, gradient boosting) on tabular + TF-IDF features; report ROC-AUC, PR-AUC and F1 on the held-out test period, with an interpretation of feature importance.

5 **Model improvement** A3

Optimise hyperparameters of the best classical model and add a text branch by **fine-tuning a pre-trained multilingual transformer** (transfer learning from `xlm-roberta-base` / `rubert-tiny2`) on lot descriptions; compare with, and ensemble into, the tabular model.

05 Type of AI task

✓ Classification | + NLP component | Regression | Clustering | Computer vision | Generative AI

Binary classification (supervised) on tabular data, with an **NLP component**: lot names and descriptions are free text in Russian and Kazakh, vectorised with TF-IDF for the baseline and, in the fine-tuning stage, encoded by a pre-trained transformer. Because positive cases are the minority (~20%), evaluation focuses on **ROC-AUC, PR-AUC and F1** rather than accuracy.

06 Dataset

6.1 Source and access

- **Source:** OWS v3 GraphQL API, `ows.goszakup.gov.kz/v3/graphql` — the official open-data interface of the public procurement portal. Access via a personal API token (valid until 13 Nov 2026). The data is public.
- **Scope:** all procurements with `publishDate` in **2024-01-01 … 2024-03-31 — 630,762 procurements**, about 35% of the 1,818,344 FY2024 procurements — and every linked record. Q1 2024 is old enough that every contract has had more than two years to reach a final status, so the outcome label is settled.
- **Collection:** `src/collect.py` — resumable `limit/after` pagination, stored as JSONL and Parquet. The snapshot is published as a GitHub release asset; the script reproduces it from scratch.

6.2 Entities

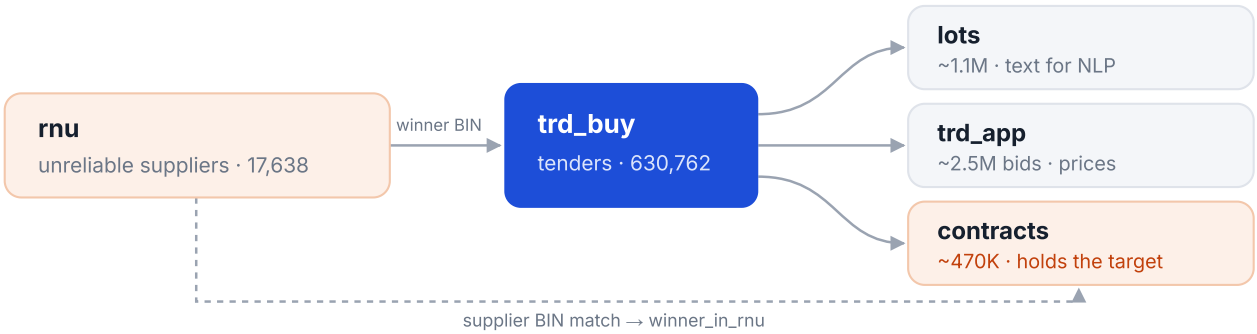


Figure 1. Linked tables. Everything hangs off the procurement id; the RNU register joins through the supplier's BIN.

TABLE	GRAIN	ROWS (Q1 2024)	KEY FIELDS
`trd_buy`	procurement	**630,762**	id, name, trade method, status, total sum, number of lots, customer BIN, winner BIN, publish / start / end dates
`lots`	lot	**~1,100,000**	procurement id, name RU/KZ, description, amount, dumping flag, KTRU code, status
`trd_app`	bid	**~2,500,000**	procurement id, supplier BIN, per-lot price and discount
`contracts`	contract	**~470,000**	procurement id, supplier BIN, status, cancellation reason, sums, planned / actual execution dates
`rnu`	supplier entry	**17,638**	supplier BIN, start / end dates in the register (all-time)

The trd_buy count is exact (fully collected). The lots estimate is extrapolated from the first 195,000 procurements collected (1.8 lots per procurement); the others from a pilot sample. Exact counts will be reported in A2.

6.3 Target variable

PRIMARY TARGET

`contract_failed` — **binary, one row per procurement**

1 if the contract from the procurement has status *Terminated* (code 350), *Awaiting termination* (345) or *Not concluded* (330); **0** if it is *Active*, *Amended* or *Executed*. Procurements without any contract record are excluded from the primary task.



Failed · 350/345/330 · **label 1** Amended · 455 Active / executed · 190/390/460 · 6% Other · 3%

Sample of the first 400 FY2024 contracts.

Auxiliary labels kept in the dataset for extended analysis:

`tender_cancelled`
~7% · procurement status Cancelled (430) or Refused (410)

`winner_in_rnu`
rare · the winner later appears in the register of unreliable suppliers

`single_source_after_failed`
~19% · re-awarded to a single source after a failed procedure

Why an observed outcome rather than the rule-based risk score: a model trained on labels produced by hand-written rules only learns those rules. Contract termination is a real, independently recorded event, so the model's quality can be measured honestly and the rules can be tested as features.

6.4 Leakage control

Only fields known at publication/award time enter the feature set: procurement and lot attributes, bids, winner identity, and the *history* of the customer and the supplier **strictly before** the procurement's publish date. Contract execution fields (`faktSum`, `faktExecDate`, status) are used for the label only. The split is by time — **train on January–February, validate/test on March 2024** — so that no future information leaks into training.

6.5 Feature groups (preliminary)

Procurement
Trade method, subject type, total sum, number of lots, planned duration, repeat flag, single-organiser flag, construction / light-industry flags, region (KATO).

Competition
Number of bidders, bids per lot, winner's discount, dumping flag, price dispersion among bids.

Text
TF-IDF (word + character n-grams) of lot names and descriptions; transformer embeddings in A3.

Supplier / customer history
Computed strictly before the publish date: number of prior contracts, share terminated, share of wins with the same customer, presence in RNU.

Rule indicators
The 31 Tender Radar indicators, re-implemented in pandas and used as input features rather than as labels.

07 Plan and deliverables

WEEKS 1–2

A1 · Problem & Dataset

This proposal; raw Q1 2024 snapshot collected and published.

WEEKS 3–4

A2 · Data Preparation

Structure, missing values, duplicates, cleaning, EDA, visualisations, features + target, time split; brief report.

WEEK 5

Midterm · Baseline

Logistic Regression + Random Forest, ROC-AUC / PR-AUC / F1, slides, justification of model choice.

AFTER MIDTERM

A3 · Tuning & Fine-tuning

LightGBM, hyperparameter optimisation, fine-tuned transformer on lot text, ensemble; experimental results.

08 Expected results

Open dataset

A public, reproducible dataset of Q1 2024 procurements with a documented outcome label.

A better classifier

ROC-AUC clearly above the rule-based score used as a baseline comparator.

Risk drivers

A ranked list of procurement characteristics most associated with contract failure — the practically useful output for an auditor.

09 Risks and mitigations

RISK	MITIGATION
API token expires 13 Nov 2026	Full snapshot collected in September; all later work is offline.
Class imbalance	PR-AUC / F1 as metrics; class weights; threshold tuning.
Label noise — termination “by mutual agreement” is not always a failure	Keep cancellation reason codes; report results on the strict subset (terminated for breach / not concluded) as a sensitivity analysis.
Text in two languages	Multilingual models (XLM-R); Kazakh-only lots handled via the Russian name, which is almost always present.
Compute for transformer fine-tuning	`rubert-tiny2` fits on CPU / Colab; LoRA for larger models.